AP CALCULUS AB

Unit 1 • Practice

# Limits and Continuity

AP-style practice • Multiple choice and free response

**How to use this practice set**

The questions follow the format of the AP Calculus AB exam. **Part A**: multiple choice, no calculator. **Part B**: multiple choice, graphing calculator required. **Part C**: free response; show your work and justify your answers, as on the exam.
Suggested timing: Part A 30 min • Part B 8 min • Part C 45 min.

## Part A • Multiple choice • No calculator

### Question 1 (1.6)

$\lim_{x\to 3} \dfrac{x^2-9}{x^2-2x-3}$ is

(A) 0
(B) $\frac{3}{2}$
(C) 3
(D) nonexistent

### Question 2 (1.6)

$\lim_{x\to 0} \dfrac{\sqrt{x+9}-3}{x}$ is

(A) 0
(B) $\frac{1}{3}$
(C) $\frac{1}{6}$
(D) 3

### Question 3 (1.8)

$\lim_{x\to 0} \dfrac{\sin(5x)}{x}$ is

(A) 5
(B) 1
(C) $\frac{1}{5}$
(D) 0

### Question 4 (1.15)

$\lim_{x\to \infty} \dfrac{4x^3-x}{2x^3+7x^2}$ is

(A) 0
(B) $\frac{4}{7}$
(C) $\infty$
(D) 2

### Question 5 (1.15)

$\lim_{x\to -\infty} \dfrac{2x}{\sqrt{x^2+5}}$ is

(A) 2
(B) $-2$
(C) 0
(D) $-\infty$

**Question 6** (1.7)

$\lim_{x \to 2^-} \dfrac{|x-2|}{x-2}$ is

(A) 1
(B) 0
(C) $-1$
(D) nonexistent

**Question 7** (1.4)

| $x$ | 0.9 | 0.99 | 0.999 | 1.001 | 1.01 | 1.1 |
|---|---|---|---|---|---|---|
| $f(x)$ | 2.71 | 2.97 | 2.997 | 3.003 | 3.03 | 3.31 |

Based on the table, which is the best estimate of $\lim_{x \to 1} f(x)$?

(A) 2.997
(B) 3
(C) 3.003
(D) The limit does not exist.

**Question 8** (1.11)

Let $f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 3x - 1, & x \geq 1. \end{cases}$ Which statement is true?

(A) $\lim_{x \to 1} f(x)$ does not exist.
(B) $\lim_{x \to 1} f(x)$ exists, but $f$ is not continuous at $x = 1$.
(C) $f$ is continuous at $x = 1$.
(D) $f(1)$ is undefined.

**Question 9** (1.13)

For what value of $k$ is $f(x) = \begin{cases} kx + 3, & x \leq 2 \\ x^2 - k, & x > 2 \end{cases}$ continuous at $x = 2$?

(A) $-\dfrac{1}{3}$
(B) $\dfrac{1}{3}$
(C) 1
(D) 7

**Question 10** (1.10)

What type of discontinuity does $g(x) = \dfrac{x^2 - 4}{x - 2}$ have at $x = 2$?

(A) Jump
(B) Infinite
(C) Removable
(D) None; $g$ is continuous at $x = 2$.

**Question 11** (1.14)

Which line or lines are vertical asymptotes of the graph of $h(x) = \dfrac{x+1}{x^2 - 1}$?

(A) $x = 1$ only
(B) $x = -1$ only
(C) $x = 1$ and $x = -1$
(D) $y = 0$ only

**Question 12** (1.8)

$\lim_{x \to 0} x^2 \cos\left(\dfrac{1}{x}\right)$ is

(A) 1
(B) 0
(C) $-1$
(D) nonexistent

**Question 13** (1.16)

The function $f$ is continuous on $[1,5]$ with $f(1) = -2$ and $f(5) = 6$. Which statement must be true?

(A) $f(c) = 8$ for some $c$ in $(1,5)$.
(B) $f(3) = 2$.
(C) $f$ is increasing on $[1,5]$.
(D) $f(c) = 4$ for some $c$ in $(1,5)$.

**Question 14** (1.15)

$\lim_{x\to\infty} \dfrac{3e^x + x^2}{e^x - 1}$ is

(A) 0
(B) 1
(C) 3
(D) $\infty$

## Part B • Multiple choice • Graphing calculator

**Question 15** (1.4)

Using a table of values or a graph, what is $\lim_{x\to 0} \dfrac{2^x - 1}{x}$, to three decimal places?

(A) 0.500
(B) 0.693
(C) 1.000
(D) 1.443

**Question 16** (1.16)

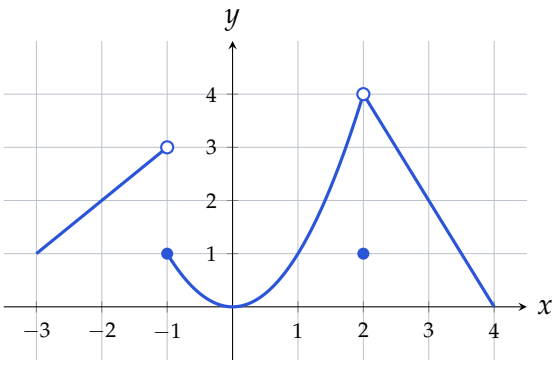
How many values of $c$ in the open interval $(0,2)$ satisfy $c^3 - 4c + 1 = 0$?

(A) None
(B) One
(C) Two
(D) Three

## Part C • Free response

**Question 17** (Graph analysis • no calculator)

The graph of the function $f$ on $[-3,4]$ is shown. It consists of a line segment for $-3 \le x < -1$, the parabola $y = x^2$ for $-1 \le x < 2$, and a line segment for $2 < x \le 4$. Also $f(-1) = 1$ and $f(2) = 1$.




(a) Find $\lim_{x\to -1^-} f(x)$ and $\lim_{x\to -1^+} f(x)$. Does $\lim_{x\to -1} f(x)$ exist? Give a reason for your answer.
(b) Find $\lim_{x\to 2} f(x)$. Is $f$ continuous at $x = 2$? Name the type of discontinuity, if any, and explain how $f(2)$ could be redefined to make $f$ continuous at $x = 2$.
(c) Find $\lim_{x\to 3} f\big(f(x)\big)$. Show the work that leads to your answer.
(d) $f(1) = 1$ and $f(3) = 2$. Explain why the Intermediate Value Theorem cannot be used to guarantee a value $c$ in $[1,3]$ with $f(c) = 1.5$. Does such a value exist? Justify.

**Question 18** (Algebraic limits and continuity • no calculator)

Let $g(x) = \begin{cases} \dfrac{x^2+x-6}{x-2}, & x < 2 \\ cx+1, & x \geq 2 \end{cases}$ where $c$ is a constant, and let $h(x) = \dfrac{x^2+x-6}{x^2-4}$.

(a) Find $\lim_{x \to 2^-} g(x)$. Show the work that leads to your answer.
(b) Find the value of $c$ for which $g$ is continuous at $x = 2$. Justify your answer using the definition of continuity.
(c) Find all values of $x$ where $h$ is not continuous, and classify each discontinuity. Justify with limits.
(d) Find $\lim_{x \to \infty} h(x)$ and $\lim_{x \to -2^+} h(x)$.

**Question 19** (Table and theorems • calculator)

The temperature $W$, in degrees Celsius, of a greenhouse is a continuous function of time $t$, in hours, for $0 \leq t \leq 10$. Selected values are given in the table.

| $t$ (hours) | 0 | 3 | 5 | 8 | 10 |
|---|---|---|---|---|---|
| $W(t)$ (°C) | 20 | 24 | 31 | 27 | 22 |

(a) Must there be a time $t$ in $[0, 10]$ at which $W(t) = 30$? Justify your answer.
(b) Explain why there must be at least two times in $[0, 10]$ at which $W(t) = 25$.
(c) Use the data to estimate the rate at which the temperature is changing at $t = 4$ hours. Show the computation and give units.
(d) A function $f$ satisfies $4x - x^2 \leq f(x) \leq x^2 - 4x + 8$ for all $x$. Find $\lim_{x \to 2} f(x)$, and name the theorem you use.

Check your answers with the answer key for this unit.